



Examining relationships between U.S. children's physical activity levels during recess, school days, and weekend days

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ABSTRACT

Objective: Physical activity opportunities, such as recess, in educational settings are vital yet only a handful of states have policies in place to ensure daily minimum recess allotment. We aim to investigate associations between physical activity during recess and other time periods to better understand the extent to which recess physical activity is related to physical activity throughout the week.

Methods: Children K-5th grade were from four schools in Little Rock, Arkansas. Children wore accelerometers on their waist to assess physical activity across the week. Chi-square and *t*-tests or Wilcoxon rank-sum tests and multi-level mixed effects generalized linear regression were used for non-normally distributed variables and accounting for repeated observations. Data collection took place from Spring 2023 through Spring 2024 and included three semesters.

Results: Children ($n = 135$, 51.9 % girls) had a median of 37.0 min of MVPA on both weekdays and weekend days. Children spent 16.7 % of recess in MVPA equaling 26.8 % of their total school MVPA, and 15.5 % of weekday MVPA. Children meeting physical activity guidelines, especially boys, had higher levels of MVPA across all time periods.

Conclusions: Recess plays a crucial role in overall physical activity, with recess MVPA showing a positive correlation with physical activity across all time periods. Further research is needed examining the effects of changes in recess policies and interventions to increase recess MVPA, which may lead to increases in children's overall physical activity.

1. Introduction

Physical activity is beneficial for children's physical and mental health, as well as cognitive and muscular development (Parrish et al., 2020; Poitras et al., 2016). A daily average of 60 min of moderate-to-vigorous physical activity (MVPA) can provide substantial health benefits, but only 25 % of U.S. children meet this recommendation (Engel et al., 2018; King et al., 2019; McKee et al., 2023). Moreover, physical activity habits established in childhood tend to persist into adulthood, influencing both current and future health outcomes (Howie et al., 2020;

King et al., 2019; McKee et al., 2023). Developing strategies to promote children's physical activity to improve their health, especially those not meeting physical activity guidelines, requires an understanding of how children's physical activity patterns vary across contexts (Jago et al., 2023).

One way to examine children's physical activity patterns is as a combination of segmented time periods, such as school day, total day and weekend days. School days are a particularly important period, children spending most of their waking weekday hours in school (Jones et al., 2020; McKee et al., 2023). School also can provide physical

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activity opportunities for nearly every child, regardless of their ability (Cooper et al., 2015; Physical Activity Alliance, 2023). However, many schools are not reaching this potential, meaning children must accumulate substantial physical activity after school and on weekends to meet the recommended 60 min per day of MVPA (Center for Disease Control, 2013; Crosnoe and Benner, 2015). Yet only around 1 in 4 children are meeting guidelines (Kretschmer et al., 2023; Michel et al., 2022). Additionally, less is known about relationships between school based physical activity and children's physical activity outside of school, such as on weekends (Bevans et al., 2010; Fairclough et al., 2012). Existing literature examining differences in weekend and weekday physical activity among children mostly focuses on characteristics such as age, sex, and weight status but less commonly for characteristics such as socioeconomic status, generally finding higher activity levels among younger children, and that boys are more active than girls (Steele et al., 2010; To et al., 2022; Wong et al., 2015). No studies to our knowledge, separate participants by those meeting/not meeting physical activity guidelines. The current study addresses two specific gaps in current literature. 1) All schools within the study are in lower socioeconomic status (SES) communities, and 2) participants have been separated by those meeting /not meeting physical activity guidelines, providing. Lastly, Arkansas law (Act 641) requires at least 40 min of recess each day, which adds novel context when discussing physical activity in schools.

Physical Education class represents one important opportunity for children to accumulate physical activity. However, Physical Education classes face numerous challenges including inadequate infrastructure, materials and equipment, low engagement, limited time, substitution with unrelated activities, lack of qualified instructors and failure to implement curriculum (Da Silva et al., 2022).

Thus, a potentially more meaningful opportunity to understand children's physical activity is recess, an unstructured daily opportunity that has been shown to increase children's total daily energy expenditure and contribute significantly to daily physical activity (Clevenger et al., 2022; Ridgers et al., 2010; Rogers et al., 2022). Although recess can contribute substantially to children's daily physical activity (Clevenger et al., 2022), physical activity levels during recess vary widely with 12 % to 50 % of total daily recess time spent in MVPA (Barenie et al., 2024; Howie et al., 2023; Lopes et al., 2006; Massey et al., 2018; Ridgers et al., 2010). Additionally, physical activity variability during recess is related to a host of different factors (Lopes et al., 2006) such as climatic conditions, recess duration, and sex where girls have been found less physically active during recess as they age, preferring different playground areas and activities compared with boys (Barenie et al., 2024).

Recess may therefore be central to understanding how children's physical activity patterns are formed. Allowing self-directed and social activities, recess may promote motivation and enjoyment of physical activity overall (Stellino et al., 2010), contributing to increased physical activity literacy and more physical activity outside of school. Despite these benefits, declines in recess time have been seen in many schools, especially those serving lower socioeconomic status (SES) areas (Brandon and Atkins, 2024; Tsai et al., 2023). In light of reductions in time allocated to recess, understanding recess physical activity and its contribution to and association with overall physical activity is needed, especially in lower SES areas where risk for inactivity is high (Barenie et al., 2024; Clevenger and Montoye, 2023; Wolfe et al., 2020).

A gap in knowledge remains concerning relationships between recess and physical activity. We therefore aim to 1) Examine and compare physical activity of elementary school children in lower SES areas, during recess, school days, weekdays and weekend days, comparing physical activity by sex and adherence to physical activity guidelines. 2) Determine the strength of associations between physical activity during recess and other time periods throughout the week.

2. Methods

2.1. Setting

This work is part of an ongoing study (Barenie et al., 2023, 2024) which explores green schoolyards and children's physical activity. This investigation was conducted in four elementary schools in Little Rock, Arkansas. All schools were in lower-income neighborhoods with limited access to walkable city parks. Data are from three observation periods across two different academic years (Spring 2023, Fall 2023, Spring 2024). Assessments were conducted twice per year. Students were invited to participate each semester, with up to three observations per child. The study was approved by the University of Arkansas for Medical Sciences IRB (274741) and participating school district. Child/parent dyads were recruited to participate through flyers, informing parents during student car-rider-lines, and by engaging with parents at school events. A parent or guardian provided written, informed consent. Children provided assent before being enrolled.

2.2. Participants

Enrollment data from the Arkansas Department of Education indicated 1300 total student enrollments of the four schools. The enrollment-weighted average for student demographics across all schools was 86 % African American, 6 % white, 6 % Hispanic/Latino, and 2 % other race or ethnicity. Of all enrolled, ~93 % were eligible for free and reduced-price lunch. Eligible children included those in grades K – 5 at each of the selected schools. Daily recess schedules varied by school. Recesses generally lasted between 15 and 20 min and were scheduled twice daily, once in midmorning and again pre/post-lunch, ensuring up to 40 min recess per day in accordance with state requirements. Recess periods included one or two grade levels.

2.3. Data collection

Student physical activity for total weekday, school day, weekend day, and recess were measured using GT9X accelerometers (ActiGraph, Pensacola, FL, USA) secured with an elastic belt around the waist. Participants were asked to wear devices for seven days, 24 h per day excepting water-based activities. Data were recorded in 15-s epochs and re-integrated into 60 s epochs to match Evenson et al. (2008) protocols, which most closely match the sample and methods in the current study (Banda et al., 2016). ACTi Life software (version 6, ACTi graph Inc) was used to process total day and school day physical activity using school schedules (Troiano et al., 2008). To identify waking wear, in-bed and out-of-bedtimes were visually identified from two independent coders, augmented by wear-time/sleep diaries when available. Discrepancies exceeding 15 min necessitated coders discussing findings until consensus was reached. Cut points (values ≥ 2296 cpm) from (Evenson et al., 2008) were used to define light, moderate and vigorous intensity physical activity (Rich et al., 2013; Wong et al., 2024). The primary physical activity variables were average minutes of physical activity per day in each intensity level analyzed as a continuous variable. Meeting physical activity guidelines was defined as having an average MVPA of at least 60 min per day over the wear period (Wong et al., 2024).

Schoolchildren recording at least one weekend day and three weekdays per observation were included, enabling comparisons of weekday vs weekend levels and durations of physical activity. Daily physical activity metrics from children who accumulated minimum wear time were averaged per observation period. Children with at least 3 valid days of weekday and one valid day of weekend wear time had average physical activity calculated by weekdays and weekends separately. Any non-wear time or partial days below minimum criteria were excluded. One observation cycle was collected per child per semester. Any child who participated in multiple semesters had their data analyzed separately for each semester in which they participated

resulting in one unique observation per semester.

After the consent/assent process, participant’s sex, age/grade and height/weight were recorded, to enable comparisons across student demographics. Accelerometers were then fitted. CDC recommended BMI percentiles were used to determine children’s weight status (Center for Disease Control, 2022). Due to ethnic homogeneity, we did not include ethnic demographic data.

2.4. Statistical analysis

To examine physical activity during recess, school day, weekday and weekend day periods, descriptive statistics for the first observation of each child (baseline), including accelerometer-measured physical activity during recess, school hours, total weekdays, and weekend days, were calculated for the total sample, by sex, and by those meeting/not meeting physical activity guidelines. To compare physical activity between sexes and those meeting guidelines during each time period, separate regression models for each time period were examined. Multi-level mixed effects generalized linear regression accounting for repeated observations within a child were used with dichotomous variables of sex or physical activity guideline adherence as the independent variable and physical activity as minutes per day as the dependent variable. Models comparing physical activity between sexes were adjusted for age and models comparing those meeting and not meeting guidelines were adjusted for age and sex. Models for moderate, vigorous, and MVPA utilized multilevel negative binomial regression to account for the count-like dispersion of the dependent variable. Finally, to determine associations between recess MVPA as the independent variable with MVPA during other time periods as the dependent variable, multilevel linear regressions were used, adjusted for age, sex, and school. Non-parametric *P*-values are presented, however means and standard deviations are presented for consistency in interpretation. All analyses were conducted in Stata/IC v 14.2 (College Station, TX, USA) with statistical significance set at $P \leq .05$.

3. Results

3.1. Sample

The sample size totaled 135 unique children, with 208 total observations. Seventy-two children had one valid observation, 53 had two observations, and 10 had three observations. Observations collected in different semesters from the same child were averaged and analyzed separately. Of these children, 51.9 % ($n = 70$) were girls and 48.2 % ($n = 65$) were boys. Eighty observations were excluded due to insufficient wear time. Of those observations excluded, 44 (55 %) were girls. The mean age was 7.7 (SD 1.8) years which did not statistically differ from those included. The average child’s age was 8.0 years (SD 1.7) at baseline assessment with an equal representation across school grades as seen in Table 1.

Thirty-three percent ($n = 44$) of the sample were with obesity. There were no statistically significant differences between boys and girls in age, grade or obesity. Twenty-three (17 %) children met physical activity guidelines. Children meeting guidelines were younger (7.3 years versus 8.2 years, $P = .020$) and had a lower percentage of girls (21.7 % vs 78.3 %, $P = .002$) compared to those not meeting guidelines. There was no difference in obesity status by sex.

3.2. Examination of physical activity data by time period

Children recorded an average of 882.4 (SD 56.4) minutes per day of waking wear-time and 5.8 (0.8) total days of wear across all observations. The baseline average daily minutes of MVPA can be seen for total days, weekdays and weekend days, school hours and recess in Table 2. At baseline, children averaged 43.2 (3.7) minutes MVPA per day across all days. Average weekday MVPA was 41.8 (23.3) daily, which was not

Table 1

Children’s weight status, grade/sex, and physical activity distributions, mean (SD) or N(%) in Little Rock, Arkansas, United States, January 2023 to May 2024.

	Total (n = 135)	Boys (n = 65; 48.1 %)	Girls (n = 70; 51.9 %)	<i>P</i> -value comparing boys and girls
Age	8.0 (1.7)	8.1 (1.8)	8.0 (1.8)	0.619
Grade (%)				0.851
Kindergarten	23 (17.0)	10 (15.4)	13 (18.6)	
1st	20 (14.8)	10 (15.4)	10 (14.3)	
2nd	22 (16.3)	11 (16.9)	11 (15.7)	
3rd	27 (20.0)	15 (23.1)	12 (17.1)	
4th	26 (19.3)	10 (15.5)	16 (22.9)	
5th	7 (12.6)	9 (13.9)	8 (11.4)	
Obese (%)	44 (32.6)	17 (26.2)	27 (38.6)	0.308
Meeting physical activity guidelines (%)	23 (17.0)	19 (27.7)	5 (7.1)	0.002

Physical activity guidelines an average of at least 60 min per day over the wear period.

P-value comparing boys and girls at baseline from chi-squared or *t*-tests.

Table 2

Children’s Physical Activity by Time Period, Mean (SD), in Little Rock, Arkansas, United States, January 2023 to May 2024.

	Recess	School Day	Total Day	Weekday	Weekend Day
Wear Time	37.8 (4.1)	420.3 (16.8)	883.1 (51.1)	891.5 (56.0)	892.3 (87.5)
Days of Wear	3.9 (0.8)	4.0 (0.7)	5.9 (0.9)	4.2 (0.64)	1.9 (0.42)
Vector Magnitude (counts per minute)	2119.1 (1016.4)	1166.2 (339.9)	1128.8 (292.2)	1123.3 (290.0)	1135.1 (401.4)
Sedentary (min/day)	10.0 (5.3)	170.6 (50.9)	380.9 (88.3)	377.1 (94.0)	385.3 (109.9)
Light (min/day)	21.1 (4.7)	228.0 (45.6)	457.7 (67.6)	471.2 (71.7)	432.2 (80.1)
Moderate (min/day)	5.2 (4.3)	17.2 (9.3)	35.7 (19.1)	34.7 (18.8)	37.2 (25.1)
Vigorous (min/day)	1.5 (2.1)	4.5 (3.8)	8.8 (7.4)	8.4 (7.3)	9.4 (10.7)
MVPA (min/day)	6.7 (5.9)	21.7 (12.4)	44.5 (25.0)	43.2 (24.5)	46.6 (33.3)

statistically different than 45.8 (32.2) minutes per weekend day across the total sample. When examined separately by sex, boys had significantly higher MVPA on weekends compared to weekdays (55.5 SD 36.7 vs 48.1 SD 27.9, $P = .048$), while there was no difference in girls. However, when the interaction between weekend or weekday status and sex was tested in a negative binomial regression, the interaction was not statistically significant ($P = .379$). Children averaged 22.1 (12.1) minutes MVPA during school hours, which was 55.6 % (18.9) of weekday MVPA. Children recorded 37.0 (4.4) minutes of recess per day, of which 16.7 % (6.3 min/day SD 5.4) was spent in MVPA. Recess MVPA accounted for 26.8 % (19.3) of total school day MVPA, and 15.5 % (11.5) of weekday MVPA. The contribution of recess MVPA did not differ by sex for school physical activity or weekday physical activity.

3.3. Physical activity data by sex

When using multilevel models for all available observations adjusting for age, comparing total daily averages between girls and boys, boys participated in higher moderate (39.6 SE 1.9 vs 29.8 SE 1.4 min/day

respectively, $P < .001$), vigorous (10.2 SE 0.9 vs 6.7 SE 0.6 min/day, $P < .001$), and combined MVPA (49.8 SE 2.6 vs 36.4 SE 1.8 min/day, $P < .001$) compared to girls as shown in Fig. 1, with all comparisons shown in Supplemental Table 2. This was consistent across time periods, except for no statistical differences between girls and boys in vigorous physical activity during school. Girls participated in more light physical activity during recess (21.2 min/day SE 0.5) compared to boys (19.7 SE 0.5, $p = .028$).

3.4. Physical activity data by physical activity guidelines

Using all available observations, physical activity in each setting is compared between those meeting and not meeting guidelines in Fig. 2. Children meeting physical activity guidelines had higher average vector magnitude, lower sedentary minutes, and higher minutes of moderate, vigorous, and combined MVPA across all settings when adjusted for age and sex. Children meeting activity guidelines showed significantly higher MVPA across the total day (71.9 SE 4.0 vs 35.5 SE 1.1 min/day, $P < .001$), school day (35.1 SE 2.3 vs 20.7 SE 0.7 min/day, $P < .001$), weekday (74.6 SE 4.8 vs 34.6 SE 1.1 min/day, $P < .001$), weekend day (91.4 SE 8.9 vs 33.3 SE 1.6 min/day, $P < .001$), and during recess (8.4 SE 1.1 vs 5.7 SE 0.5 min/day, $P = .008$) as shown in Fig. 2 with all comparisons shown in Supplemental Table 2. There was no difference in light activity between those meeting and not meeting guidelines, except during school, where those meeting guidelines had more light physical activity (238.8 SE 6.7 min/day) compared to those not meeting guidelines (222.6 SE 3.3, $P = .027$).

3.5. Associations between recess physical activity and other time periods

Minutes of MVPA during recess, were positively associated with total MVPA ($b = 0.03$ min/day, 95 %CI: 0.02, 0.04, $P < .001$), school MVPA ($b = 0.04$ min/day, 95 %CI: 0.03, 0.05, $P < .001$), and weekday MVPA ($b = 0.03$ min/day, 95 %CI: 0.03, 0.05, $P < .001$) when adjusting for age and sex. We found no association between recess MVPA with weekend MVPA ($b = 0.01$ min/day, 95 %CI: -0.01, 0.03, $P = .172$). Similarly other intensities of physical activity including light, moderate, vigorous, and vector magnitude during recess were positively associated with light, moderate, vigorous, and vector magnitude during school, total day and weekdays respectively which can be seen in Supplemental Table 3.

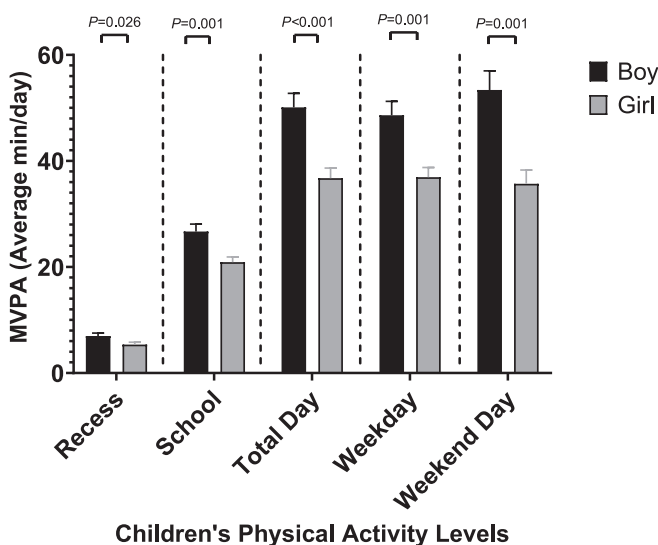


Fig. 1. Children's accumulated moderate to vigorous physical activity, estimated marginal means (SE), separated by time period and sex, in Little Rock, Arkansas, United States, January 2023 to May 2024.

Mean = +. P-values from multilevel negative binomial regressions adjusted for age, comparing boys to girls within each time period.

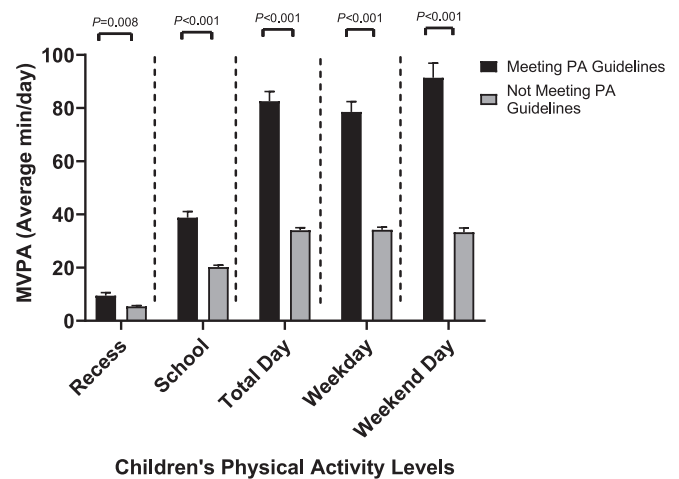


Fig. 2. Distribution of children meeting vs not meeting moderate to vigorous physical activity guidelines at different time periods, estimated marginal means (SE), in Little Rock, Arkansas, United States, January 2023 to May 2024. P-values from multilevel negative binomial regressions adjusted for age and sex, comparing those meeting and not meeting physical activity guidelines within each time period.

Moderate physical activity during recess and recess vector magnitude were positively associated with weekend moderate and vector magnitude, however, similar to MVPA, light and vigorous physical activity were not associated with weekend light and vigorous physical activity.

4. Discussion

The current study assessed the contribution of recess MVPA to daily MVPA across time periods, comparing physical activity by sex and children meeting/not meeting physical activity guidelines. We examined associations between recess MVPA and school day/weekday/weekend MVPA. Recess MVPA contributed significantly to total MVPA during school and on weekdays and was positively associated with school day and weekday MVPA. Recess emerged as a potential indicator of physical activity, and key contributor to weekday MVPA, though unequally distributed by sex.

Girls accumulated less MVPA than boys during all time periods which aligns with existing literature (Barenie et al., 2024; Graham et al., 2021; Lemberg et al., 2023; Ridgers et al., 2012; Tanaka et al., 2019). Differences are attributable to social contexts, structure of the environment, and behavioral differences (Barenie et al., 2024; Wong et al., 2024). It may also indicate that physical activity opportunities better suit boys (Barenie et al., 2024), particularly weekend activities (Poulos et al., 2024; Pulido Sánchez and Iglesias Gallego, 2021). Girls showed no increase in MVPA on weekend days compared with boys whose MVPA increased by 15 %, potentially due to a lack of physical activity options that fit well with girl's preferences (Kretschmer et al., 2023; Massey et al., 2020; Pawlowski et al., 2014; Weisgram, 2022). Further attention to girls and children not meeting guidelines is needed.

Only 17 % of our sample met MVPA guidelines, compared to approximately 25 % of children across the United States (Engel et al., 2018; King et al., 2019; McKee et al., 2023), which may be related to SES. Socioeconomic indicators (education, occupation, and income) have been identified as possible determinants of children's physical activity (Ziegeldorf et al., 2024). As mentioned, 93 % of children within our study schools qualify for free/reduce priced lunches. These children may be facing additional barriers and fewer physical activity opportunities (Hankonen et al., 2017).

Vulnerable populations in lower SES communities remain under-represented in the existing research (Bürge et al., 2016; Cosgrove and

Castelli, 2018; Drenowatz et al., 2010). Continued research on these communities is crucial. Also, future research examining highly active children within these communities may enhance our understanding of physical activity opportunities and barriers. Children meeting physical activity guidelines showed higher MVPA across all time periods. During school, they accumulated 70 % more MVPA than those not meeting guidelines and nearly half an hour more MVPA on weekdays. Lastly, they accumulated nearly an hour more MVPA per weekend day, further supporting the need to understand which activities most contribute to highly active children.

These findings provide interesting comparisons to existing research. Children meeting physical activity guidelines in our study were substantially more active on weekends. Boys in our sample accumulated 15 % more physical activity on weekends, contrasting existing studies that have shown lower activity levels on weekends. Children not meeting physical activity guidelines were similarly active on weekends. We must consider the unique cultural context as a potential mechanism behind these findings. It may be the case that higher recess allotment is related to increases in weekend physical activity. This provides a meaningful direction for future research to examine weekend physical activity opportunities in similar communities, and potential impacts of recess allotment on physical activity across the whole week. Future research may compare physical activity across the week among children who receive different amounts of daily recess. We additionally intend to measure children's reported participation in organized physical activity, to better understand the specific weekend activities contributing to higher physical activity levels outside of school.

The study has several strengths and limitations. Strengths include a focus on physical activity across a one-week period, accelerometry-based measurement of physical activity, and a balanced grade/sex distribution among participants. The current data is part of an ongoing study. Longitudinal findings will increase our ability to identify trends over time within/between participants including adequate power to test interactions between demographics and time periods. Limitations include a cross-sectional design, precluding causal inferences about recess MVPA having a direct impact on MVPA outside of school. A lack of situational context limits the explanation of specific behaviors. Additionally, not including Physical Education as a covariate limits our understanding of factors impacting weekly physical activity. However, AR Code § 6-16-132 (2024) currently requires only 40 min per week of Physical Education Training and Instruction. As such, Physical Education may not provide an indication of daily physical activity within our current sample and as such, was outside the scope of the current study. Future studies may include details about children's weekly Physical Education.

5. Conclusions

The current study reinforces the school's critical role in supporting physical activity opportunities for less active children living in lower SES communities. Recess is an especially important opportunity for children not accumulating physical activity at other times in the week. Although it is true that recess alone is not sufficient for children to meet daily MVPA guidelines, it may be the case that minimum daily recess is essential for children who are inactive outside of school. Schools are positioned to provide activity opportunities for all children. But they need additional support and may be nearly maxed out already in their ability to facilitate physical activity. To reduce physical activity disparities, schools in lower SES communities must be empowered to protect recess time as a critical contributor to children's overall physical activity. Equally important is the need to gain a deeper understanding in these specific communities, of common factors that may indicate higher levels of physical activity. This will require continued analysis of physical activity in other contexts such as Physical Education class and outside of school, weekends in particular.

CRedit authorship contribution statement

Christopher M. Murphy: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Erin K. Howie:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Matthew J. Barenie:** Writing – review & editing, Visualization, Methodology, Investigation. **Kari A. Weber:** Writing – review & editing, Investigation. **Michael R. Thomsen:** Writing – review & editing, Software, Project administration, Investigation, Funding acquisition.

Ethics approval and consent to participate

Ethical approval for this study has been received by the University of Arkansas for Medical Sciences Institutional Review Board (Protocol #274741). All methods were carried out in accordance with relevant guidelines and regulations.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pmedr.2025.103120>.

Data availability

This article is part of a broader and ongoing four-year study. De-identified data will be made available in a public repository at the end of the study period. The data that has been used is confidential.

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